

Prescribed-front Icepack experiment at EKaS: implementation, synthetic results and remaining problems

Icepack-Sediment

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Result status. Both synthetic experiments completed a common 25-year spinup and the 1953–2021 transient in Firedrake/UW Icepack. At the comparison gate, the advancing case is 67.83 m thicker and 515.54 m/year slower than the fixed-front control in 2021. These results validate a numerical experiment; they are not EKaS predictions or agreement with the publication. A regional run could not start because prepared regional inputs are missing. A geometry audit also rejects the tested straight corridor.

The repository is public. The exact `AGENTS.md` and academic style guide were copied from OMP-OOI main. OMP-OOI has no standalone plotting style file; the implementation reuses the `rcParams` block from its energy-likelihood analysis and 220 dpi exports. Source commits and hashes are recorded in `template_provenance.json`. The implementation was introduced in pull request 1; these validation and template updates are submitted from the feature branch for review.

1 Relationship to Figure 13

Dachauer et al. (2026), doi:[10.1017/jog.2026.10172](https://doi.org/10.1017/jog.2026.10172), Figure 13, compares prescribed front advance with a fixed-front control. The gate here lies 1 km upstream of the nominal 1953 front. The paper initializes from 2021 geometry, inverts fluidity and slipperiness, relaxes for two years, re-inverts, truncates at the 1953 front and relaxes for 25 years. This code implements truncation, common relaxation and paired transients, but not the inversions. It does not reconstruct a measured 1953 surface. The front offsets are 0, 960, 1400 and 1820 m at 1953, 1985, 2007 and 2021. They integrate rounded published advance rates rather than mapped front lines. Editable CSV observations drive monotone PCHIP; piecewise-linear interpolation is available to follow the paper’s interpolation convention.

2 Physics and moving domain

Icepack `IceStream` supplies SSA dynamics, Glen exponent 3, ocean traction and hydrostatic grounded/floating surface geometry. Regional forcing is intended to use a fixed MAR 2021 SMB field and observed upstream velocity and thickness. Those fields have not yet been prepared. Synthetic tests use zero SMB, a sloping synthetic bed and surface, and prescribed inflow. With speed v , $k = C_{\text{Ua}}^{-1/3}$ and effective pressure N , sliding follows

$$t_b = \frac{\mu N k v^{1/3}}{\mu N + k v^{1/3}}, \quad N = \max\{\rho_i g H - \rho_w g \max(-z_b, 0), 0\}, \quad z_b = s - H.$$

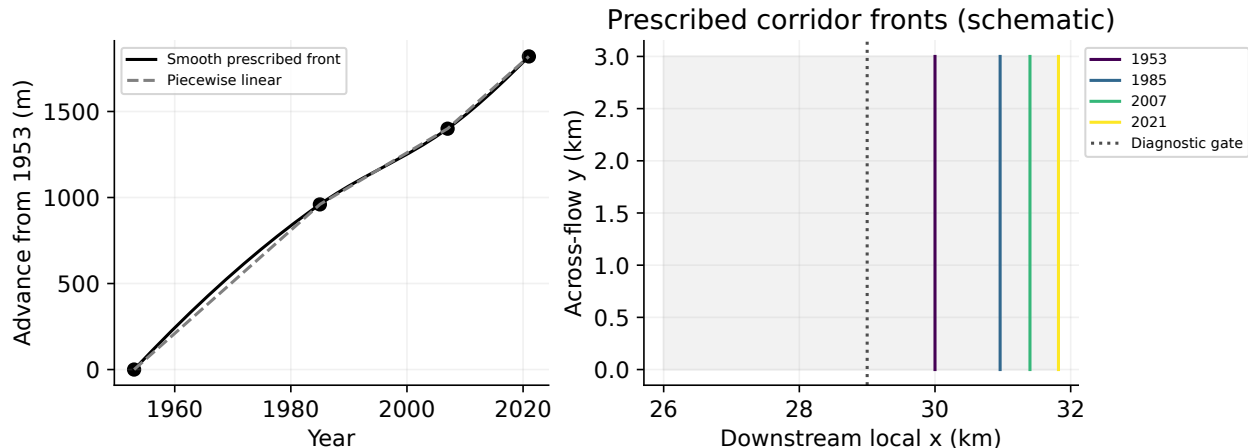


Figure 1: Approximate prescribed chronology and schematic model front positions. The lines are not mapped historical terminus boundaries.

Floating ice has zero basal traction. A 16-point speed quadrature integrates the traction to obtain Icepack’s dissipation potential; speed is regularized at 0.001 m/year. The default $\mu = 0.5$, uniform fluidity $A = 10$ in MPa–m–year units and supplied/prior friction are assumptions, not recovered Ua parameters. Side walls allow tangential sliding with configurable drag.

Each timestep builds an identically ordered triangular corridor mesh at the prescribed length. Thickness transfer uses the old/new length ratio to preserve volume; implicit continuity uses physical velocity minus mesh velocity. This first-order ALE split creates no arbitrary new ice slab and records discharge relative to the moving boundary. It is a straight constant-width domain, not the curved glacier catchment. CG1 sampling uses actual triangle connectivity. Serial execution is required. Negative or nonfinite thickness aborts rather than being clipped. Fresh meshes and form construction make long runs expensive.

3 Executed simulations and checks

The local Docker image `vdv-ice-shelf` supplies working Firedrake and UW Icepack. The full synthetic pair used a 12 by 4 mesh, $\Delta t = 0.25$ year, 25-year spinup and 68-year transient. The retained production potential uses 16-point speed quadrature and Icepack’s spatial degree 2 for CG1 spaces.

Quantity	Fixed	Advancing
Minimum thickness over run (m)	431.82	408.40
Maximum relative mass residual	3.91×10^{-16}	4.50×10^{-16}
2021 front offset (m)	0	1820
2021 gate thickness (m)	482.57	550.41
2021 gate speed (m/year)	3173.47	2657.93
2021 gate flux (Gt/year)	4.21292	4.02450

Table 1: Actual full-duration synthetic outputs. Mass residual is the continuity volume error divided by volume; closure alone does not establish positivity or accuracy.

The validator passed identical input/configuration and initial-state hashes, equal post-spinup fields, stationary control front, prescribed advancing front, positive thickness, mass balance and the expected qualitative slowdown/thickening. Advancing-minus-control flux is -0.18842 Gt/year. Five unit tests passed for chronology, friction limits and raster preprocessing (grounded/floating thickness, SMB units, velocity rotation and nodata rejection). Firedrake checks passed CG1 sampling, conservative transfer and 16 traction-derivative cases. The executed notebook retains outputs and imports the core implementation.

An experimental analytic potential also completed the full pair: final gate thickening was 67.8368 m and slowdown 515.5849 m/year. Short tests with spatial degrees 12 and 24 differed by less than 5×10^{-13} in gate speed/flux. The analytic variant was not retained: no controlled runtime benefit was established. Its recorded results describe that experiment, not an additional production mode. Neither these checks nor mass closure establish mesh or timestep convergence.

4 Problems found

A one-year timestep failed during the common spinup at elapsed year 4 with negative thickness, before any advancing-front forcing. Both the original and experimental analytic potential showed this failure; the latest saved analytic failure had minimum thickness -7535.07 m. Partial histories and failure JSON are retained locally. The quarter-year pair stayed positive, but halving that timestep and refining the mesh remain necessary. CG1 transport and split stress/continuity updates do not guarantee positivity.

Cached BedMachine Greenland v6 and a modern flowline enabled a geometry probe. The candidate 30 km by 3 km straight corridor used the modern front minus a 2 km fjord extension and the approximate 1820 m advance to estimate the initial front. Of sampled initial grid points, 25.94% fell outside the ice mask and 23.19% had nonpositive thickness. These are sampled-point fractions, not exact area fractions. The candidate cannot be used as positive-thickness regional input. This audit does not rule out every alternative corridor; a curved ice-conforming domain or a verified narrower trunk is needed.

The regional invocation failed with missing `data/processed/ekas.npz`. Cached summer-2023 MEaSUREs velocities cannot silently replace annual-2021 velocity. MAR 2021 SMB, verified historical boundaries, calibrated friction and fluidity and the author’s corrected/kriged bed remain unavailable to this run. BedMachine v6 is an available substitute, not the paper’s exact corrected bed. The plotting environment lacks Rasterio; all five preprocessing tests instead passed in the separate `ekas-figure` environment. Default Python lacks Firedrake/Icepack; the Docker checks establish availability only in that image.

5 Reproduction and next work

The README documents installation, input preparation, paired runs, validation, figure generation and report compilation. Raw regional data and simulation archives are ignored by git. Source information and units are documented in `data/README.md`; small plots and validation summaries are committed. The figure script rejects incomplete simulations and explicitly labels synthetic results. Historical panels remain unavailable rather than being filled with invented model output.

Prepare valid regional geometry and 2021 forcing/velocity, calibrate the stress balance, replace rounded offsets with mapped boundaries and assess the corrected bed before interpreting the Figure 13 comparison. Then check spatial/temporal convergence and compare flux, velocity and dynamic

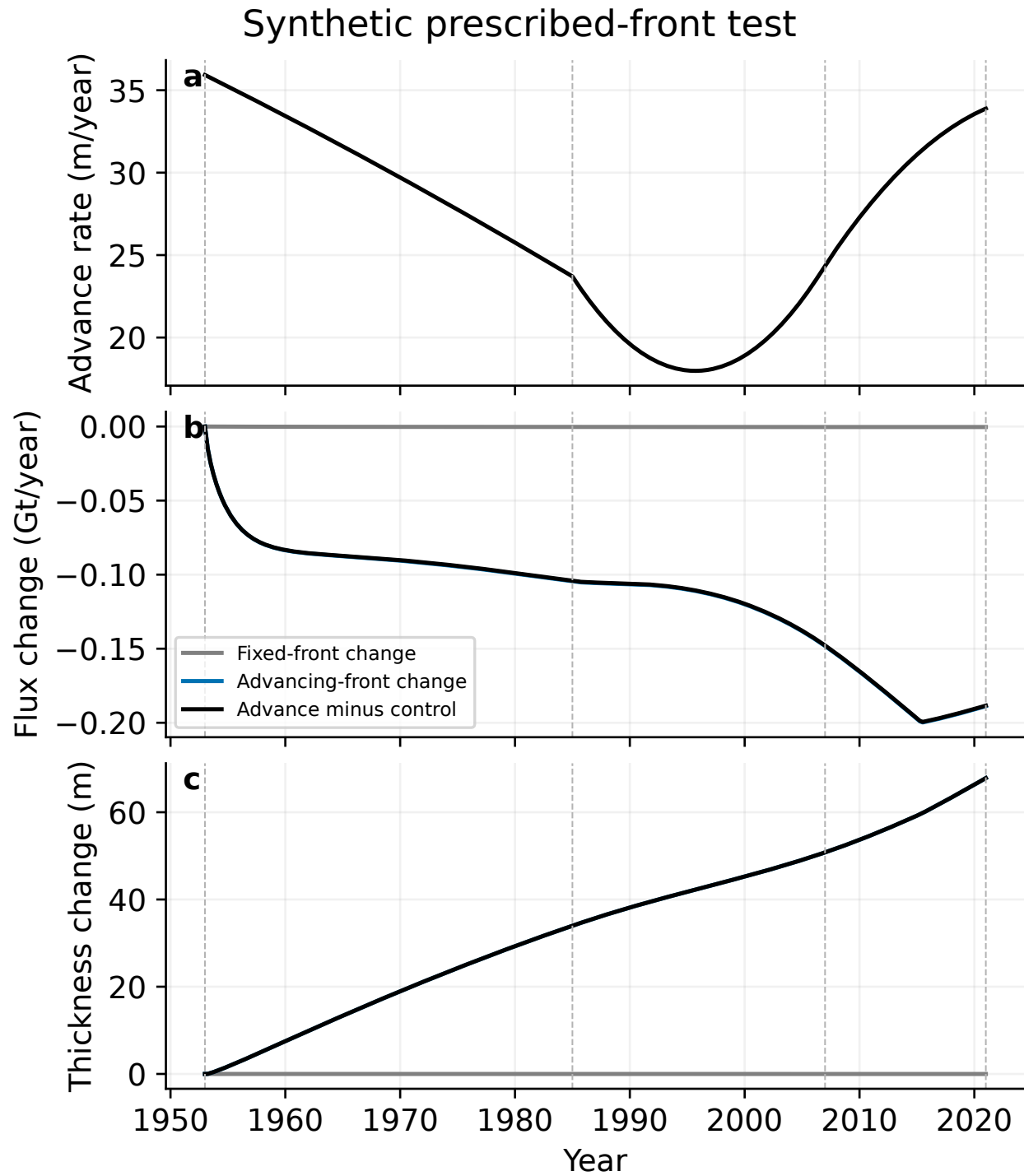


Figure 2: Executed synthetic gate comparison. The data and magnitudes are not calibrated EKaS results.

thickening. Main differences from Ua include the trunk boundary conditions, missing inversion/re-inversion, uniform fluidity, approximate fronts, stock bed and first-order corridor ALE transport. Their influence on quantitative agreement remains unresolved. Explicit sediment thickness and bank geometry are the planned next step after the geometric experiment is established.

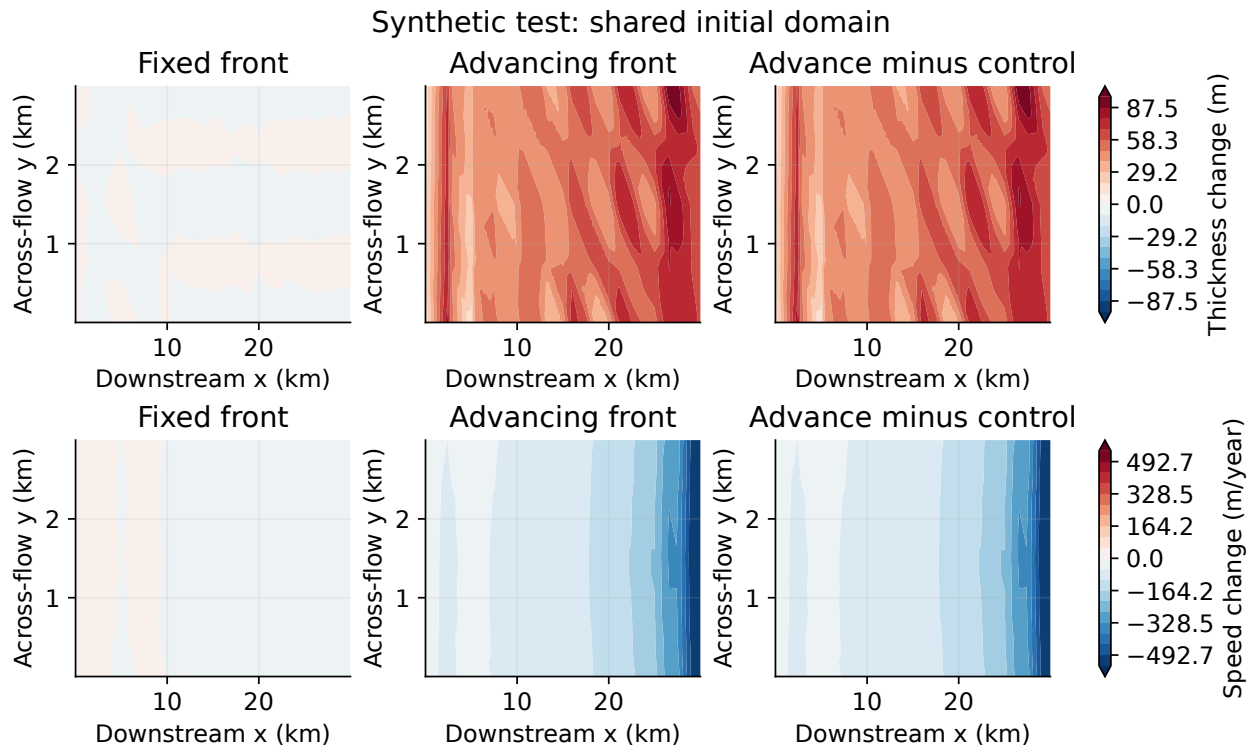


Figure 3: Synthetic spatial changes relative to the common post-spinup state and advancing-minus-control differences on the initial domain.

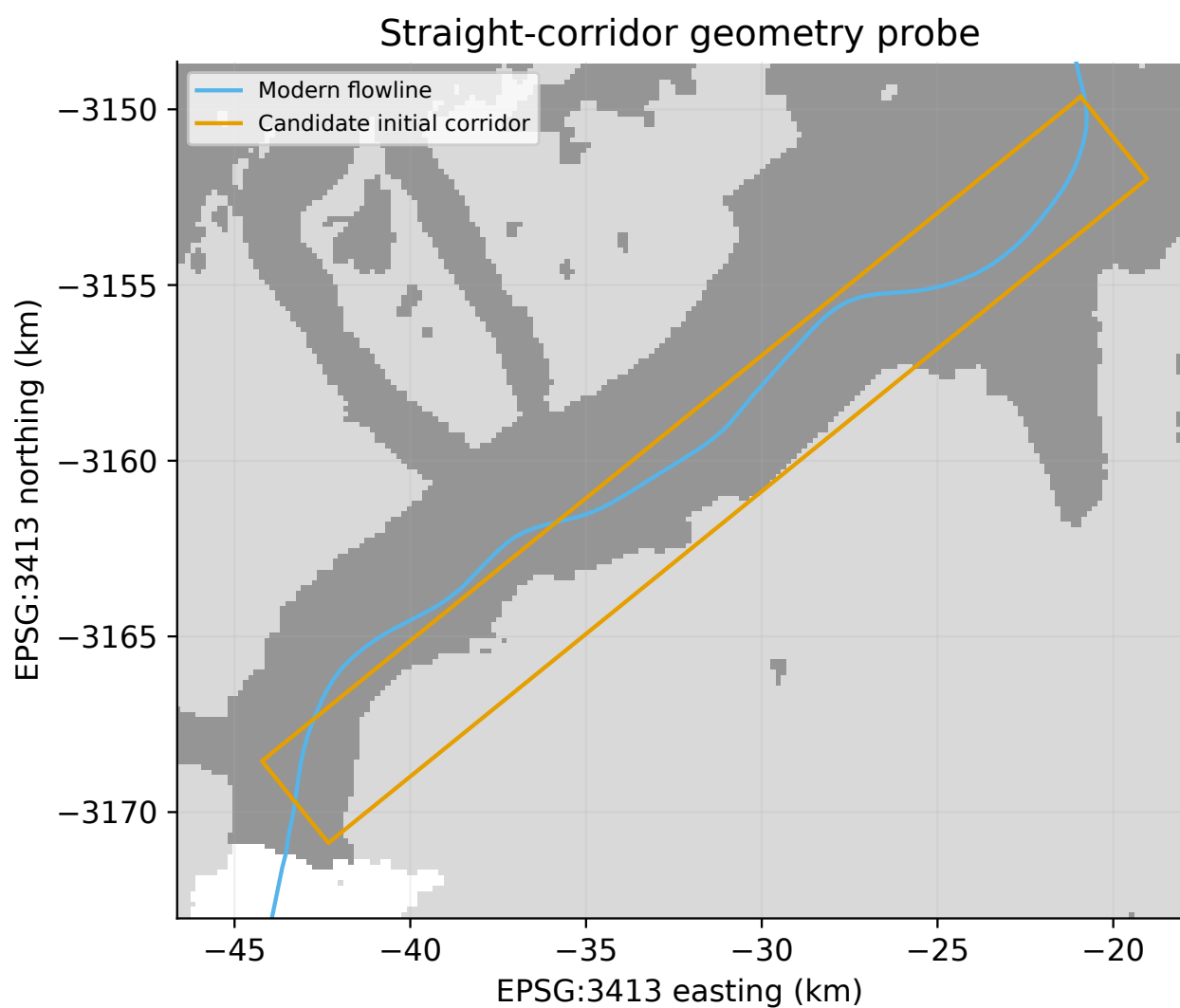


Figure 4: Geometry probe using cached BedMachine v6 and a modern flowline. The candidate initial front is approximate, not a surveyed 1953 boundary.